Approach & Assumptions

Project Approach

This project simulates and analyzes a large-scale insurance portfolio to understand premium revenue, claim patterns, and overall portfolio profitability.

The analysis was performed in three main stages:

1. Data Simulation

A synthetic dataset was generated using Python in Jupyter Notebook based on the business rules provided in the assignment. The dataset includes approximately 1 million insurance policies sold during 2024.

Two tables were created:

Policy Sales Data

Contains information about:

- Customer ID
- Vehicle ID
- Vehicle value
- Policy tenure
- Premium amount
- Policy purchase date
- Policy start and end dates

Claims Data

Contains:

- Claim ID
- Customer ID
- Vehicle ID
- Claim amount
- Claim date
- Claim type

The datasets were exported as **CSV files**.

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2. Data Validation (Python – Jupyter Notebook)

Before performing the analysis, several data validation checks were performed in Jupyter Notebook to ensure the generated data followed the business assumptions and contained no inconsistencies.

The validation process included:

- Verifying the total number of policies generated
- Checking the policy tenure distribution (20%, 30%, 40%, 10%)
- Ensuring policy purchase dates were within the range of January 1 to December 31, 2024
- Confirming that premium values matched the rule of ₹100 per year of policy tenure
- Validating that policy start dates were exactly 365 days after the purchase date
- Checking that claim amounts matched 10% of the vehicle value
- Confirming that there were no missing values or incorrect data types

These validation steps ensured that the simulated dataset was consistent with the assignment's business rules and suitable for analysis.

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3. Data Storage & Processing (SQL)

After validation, the datasets were loaded into a MySQL database.

SQL queries were then used to perform analytical calculations, including:

- Total premium collected
- Total claim costs
- Claim frequency
- Average claim amount
- Monthly claim trends
- Policy distribution by tenure
- Loss ratio analysis
- Portfolio profitability analysis
- Potential future claim liability
- Earned premium calculation

SQL allowed efficient analysis of the large dataset containing nearly one million records.

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4. Business Analysis & Insights

The results from SQL queries were interpreted to generate business insights regarding:

- portfolio revenue

- claim patterns
- policy profitability
- financial sustainability
- long-term risk exposure

These insights simulate the type of insurance portfolio analysis typically performed by business intelligence and risk analytics teams.

Assumptions Used in the Simulation

The dataset was generated based on the following business assumptions.

Policy Sales Assumptions

- 1,000,000 customers purchased insurance policies in 2024
- Policies were sold evenly across all days of the year
- Policy tenure distribution:

Tenure	Distribution
1 year	20%
2 years	30%
3 years	40%
4 years	10%

- Vehicle values were assumed to range between ₹100,000 and ₹300,000
- The premium is ₹100 per year of policy tenure
- The policy start date is exactly 365 days after the purchase date
- The policy end date is calculated based on policy tenure

Claims Assumptions

Claims were simulated based on two scenarios.

2025 Claims

- 30% of vehicles purchased on the 7th, 14th, 21st, and 28th of each month filed claims
- The claim date is the policy start date

2026 Claims

- 10% of vehicles with 4-year policy tenure filed claims
- Claims were distributed evenly between January 1 and February 28, 2026